

Metrics that matter

Simulation without metrics is hope

Pass rate, coverage, escapes

- Pass rate tells you how many tests succeed right now, one flaky fail blocks confidence
- Coverage percent tells you how much of the intended space you exercised
- Bug escapes are the painful teacher

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / Verification metrics board

INTERACTIVE TOOL

Verification metrics board

Document-aid literacy: **pass rate**, **coverage %**, and **bug escape** as a closure dashboard (concept). Starter: all bars met — board HEALTHY.

Starter example: pass 98%, coverage 95%, escapes 0 — board HEALTHY.

Load starter example

Challenges 0 / 22 cleared

Quiz: pass rate: Pass rate tracks...

☐ the fraction of regression jobs that pass versus a bar

☐ only synthesis area

☐ GTKWave cursor count

☐ Makefile PHONY targets

Show hint

Check

Next

Idle

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

Core ideas

pass rate

Regression green % vs bar.

coverage %

Plan hit rate vs goal.

bug escape

Late / field sev-1 count.

HEALTHY

Every metric meets its bar.

Lab

SCENARIO

all good

Load preset

Apply level

Scan board

Demo escape

Explain

Reset

Board HEALTHY — pass + coverage + escapes meet bars

Lab starter

Real Verilator practice

- In Track A, run a small regression, even two tests, and record pass/fail in notes
- If you have a coverage switch or simple checklist from EXAMPLES
- Name one hypothetical escape and what test would catch it
- This is literacy, not a full coverage closure project

Pitfalls to watch

- Do not celebrate hundred percent pass rate with zero coverage depth
- They teach the vocabulary
- And do not sign off when status is BLOCKED; fix the escape class or add tests first

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, reach HEALTHY on at least one scenario and explain BLOCKED on another
- In Track A, write three numbers or notes: passes, one coverage gap, one escape risk
- When you are ready, take the short quiz, then continue to the offline example